# POWER BI DASHBOARD REPORT

## Vendor Performance Dashboard

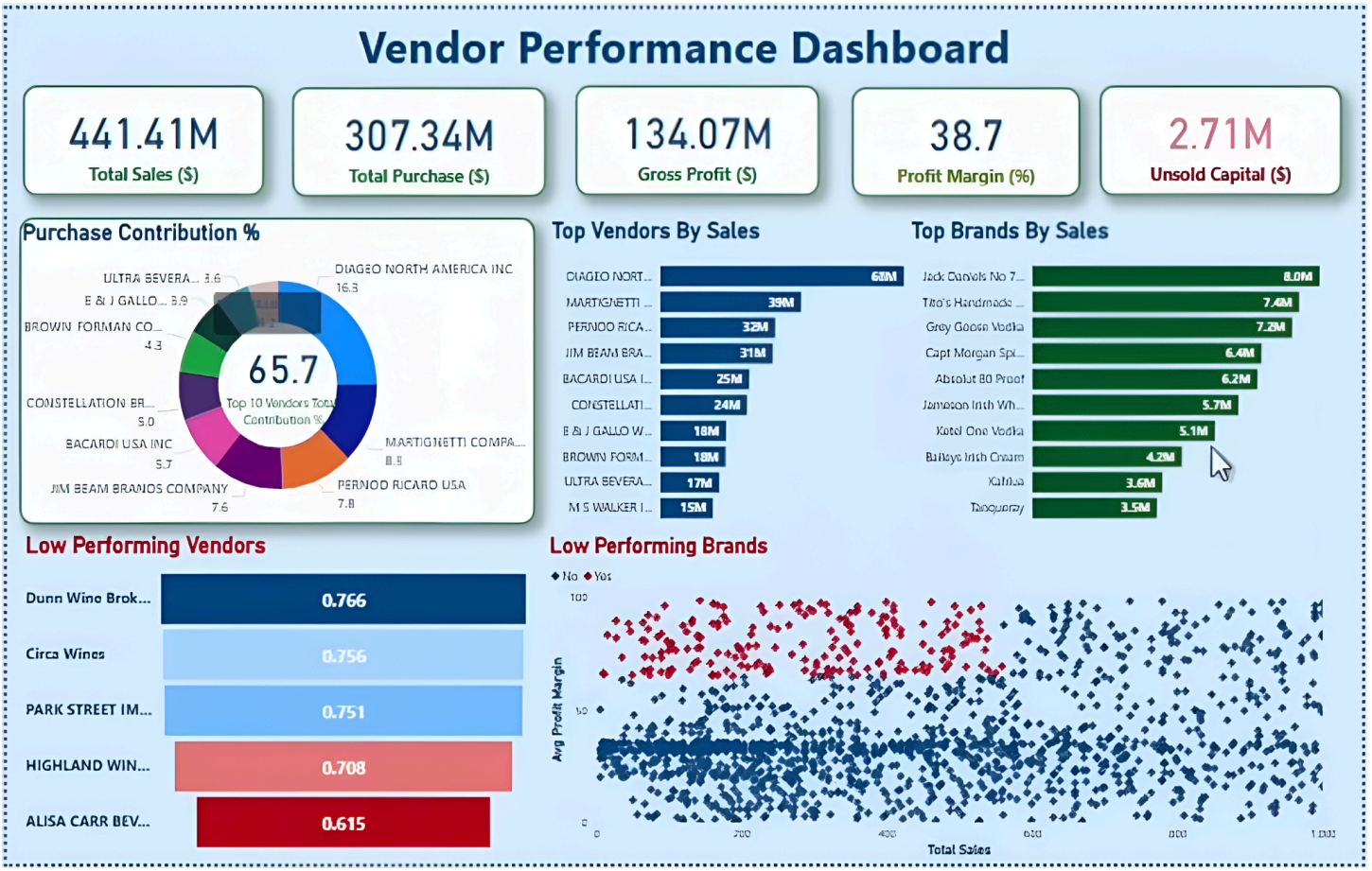
**Interactive Business Intelligence Dashboard for Procurement, Sales, and Inventory Analysis**

### Made By

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### Tools Used

Power BI, SQL (SQLite), PythonDAX



# 1. Dashboard Overview

The **Vendor Performance Dashboard** is an interactive Power BI dashboard developed to monitor vendor performance, sales, profitability, procurement efficiency, and inventory utilization. It provides business users with a centralized platform to analyze key performance indicators (KPIs) and identify opportunities for improving procurement strategies and operational efficiency.

The dashboard transforms complex analytical results into easy-to-understand visualizations, enabling faster and more informed business decisions.

# 2. Dashboard Objectives

The primary objectives of the dashboard are to:

- Monitor overall sales, purchases, and profitability.
- Identify the top-performing vendors and brands.
- Analyze vendor contribution to total purchases.
- Detects low-performing vendors and brands requiring attention.
- Monitor inventory utilization through unsold capital.
- Support procurement and pricing decisions using interactive visualizations.

# 3. Data Source

The dashboard is built using the **Vendor Sales Summary** dataset generated during the data analysis phase.

The dataset is created by integrating four business datasets:

- **Purchases** – Vendor purchase transactions
- **Sales** – Product sales transactions
- **Purchase Prices** – Actual product pricing information
- **Vendor Invoice** – Freight cost details

The integrated dataset includes calculated business metrics such as:

- Gross Profit
- Profit Margin
- Stock Turnover
- Sales to Purchase Ratio

These metrics serve as the foundation for all dashboard visualizations.

# 4. Dashboard Architecture

The dashboard development follows the workflow below:

Raw CSV Files

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SQLite Database

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SQL Data Integration

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Data Cleaning & Feature Engineering

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Vendor Sales Summary Dataset

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Power BI Data Model

⇓

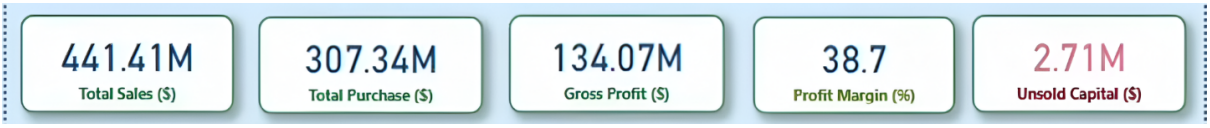
Interactive Dashboard

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Business Decision Making

# 5. KPI Cards

The dashboard contains the following KPI cards:




1. **Total Sales**

   Displays the total revenue generated from product sales.

   **Purpose:** Measures overall business performance.

2. **Total Purchase**

   Represents the total procurement expenditure incurred for purchasing inventory.

   **Purpose:** Helps evaluate procurement costs.

3. **Gross Profit**

   Displays the difference between total sales revenue and purchase costs.

   **Formula :**

   $$\text{Gross Profit} = \text{Total Sales} - \text{Total Purchase}$$

   **Purpose :** Measures business profitability.

4. **Profit Margin**

   Represents the percentage of revenue retained as profit.

   **Formula**

   $$\text{Profit Margin} = (\text{Gross Profit} / \text{Total Sales}) \times 100$$

   **Purpose :** Evaluates profitability efficiency.

5. **Unsold Capital**

   Shows the monetary value of inventory that remains unsold.

   **Purpose :** Identifies capital blocked in slow-moving inventory.

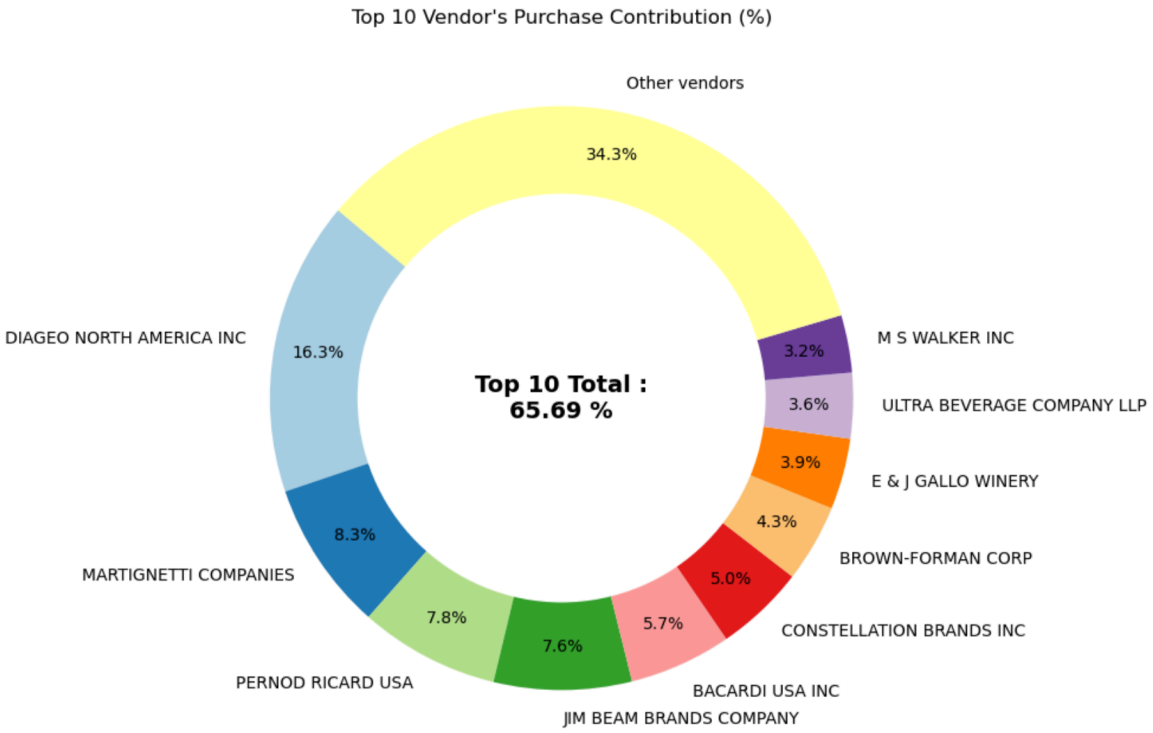
# 6. Explanation of Every Visual

## 1. Purchase Contribution %

This donut chart shows the contribution of the top vendors toward total purchases.

**Insight**

- Top vendors contribute the majority of procurement expenditure.
- Heavy dependence on a few vendors may increase supply chain risk.

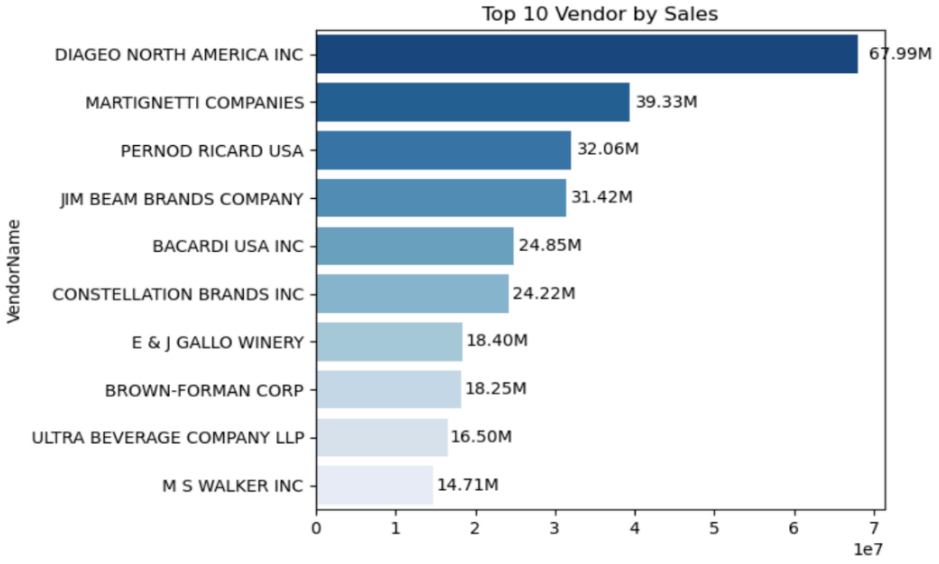



## 2. Top Vendors by Sales

This bar chart ranks vendors based on total sales revenue.

**Insight**

- Identifies the highest revenue-generating vendors.
- Helps prioritize strategic vendor relationships.

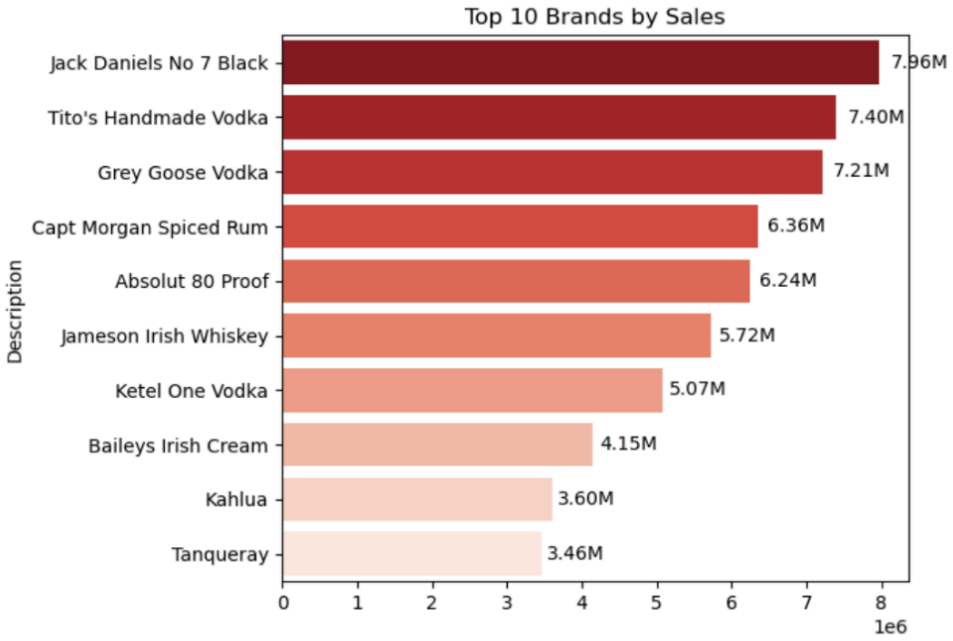



## 3. Top Brands by Sales

This visualization displays the highest-selling product brands.

**Insight**

- Identifies products with strong market demand.
- Supports inventory planning and marketing strategies.

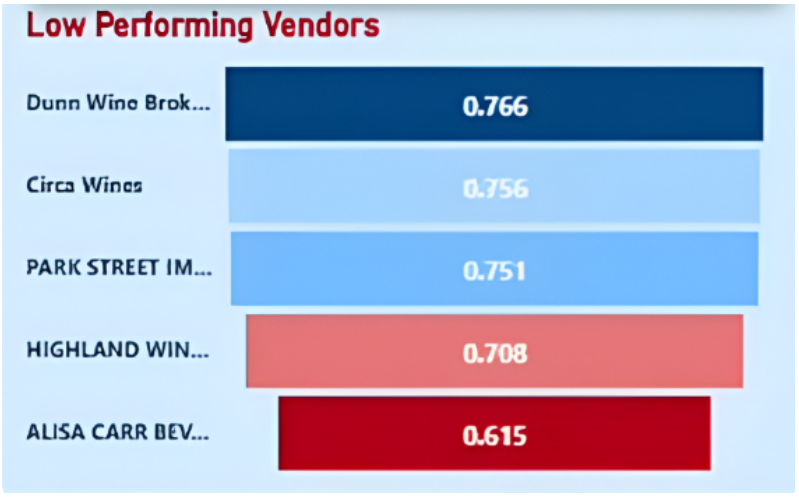


## 4. Low Performing Vendors

This chart highlights vendors generating relatively low sales.

**Insight**

- Indicates vendors requiring pricing improvements or marketing support.
- Helps management review procurement decisions.

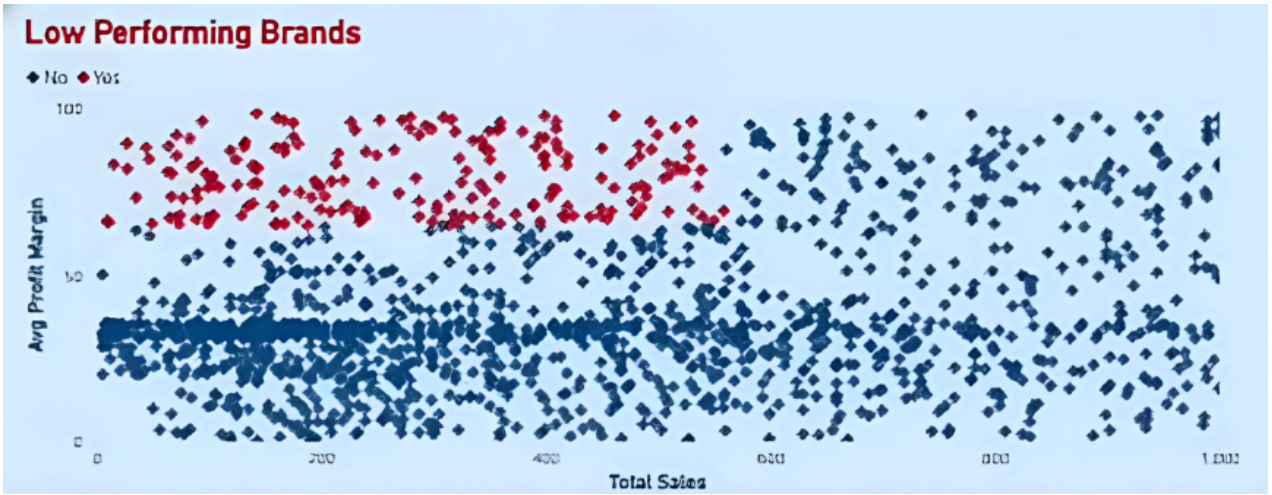



## 5. Low Performing Brands

The scatter plot identifies brands with low sales performance.

**Insight**

- Reveals brands that may require promotional campaigns, price optimization, or discontinuation.



# 7. Interactive Features (Filters / Slicers)

The dashboard provides interactive filters that allow users to dynamically explore the data.

Available filters include:

- Vendor Name
- Brand
- Product Volume
- Purchase Price
- Sales Amount

These slicers enable users to analyze specific vendors or products without modifying the underlying dataset.

# 8. Key Insights from Dashboard

The dashboard reveals several important business insights:

- A small number of vendors contribute the majority of procurement expenditure, indicating vendor dependency.
- Several brands generate high profit margins but relatively low sales, suggesting opportunities for pricing optimization or promotional campaigns.
- Significant capital is tied up in slow-moving inventory, reducing cash flow efficiency.
- Bulk purchasing leads to lower unit costs, improving procurement efficiency.
- High-performing vendors generate greater sales volume, whereas low-performing vendors generally maintain higher profit margins.

# 9. Business Decisions Supported

The dashboard supports strategic business decisions by enabling organizations to:

- Optimize vendor selection and procurement planning.
- Reduce dependency on a small number of suppliers.
- Improve inventory management through better stock monitoring.
- Identify high performing and underperforming vendors.
- Develop pricing and promotional strategies for low-selling brands.

- Improve profitability through data driven purchasing decisions.

# 10. Conclusion

The Vendor Performance Dashboard provides an interactive and comprehensive view of procurement, sales, profitability, and inventory performance. By consolidating multiple business datasets into meaningful visualizations, the dashboard enables users to monitor key performance indicators and identify operational improvement opportunities.

The insights generated support informed decision making in vendor management, inventory optimization, pricing strategy, and procurement planning. Overall, the dashboard serves as an effective business intelligence tool that transforms analytical results into actionable information, helping organizations improve efficiency, reduce operational risks, and enhance long-term profitability.